

PEDAGOGY OF PHYSICAL SCIENCE - I

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UNIT - 1

AIMS AND OBJECTIVES OF TEACHING PHYSICAL SCIENCE

1.1 Physical Science : Meaning

Meaning

Physical science is the study of inanimate natural objects including Physics, Chemistry, Astronomy, Earth Sciences and related subjects. It deals with the study of nature and properties of non-living matter and energy. There is a strong link between Physics and Chemistry. Structure of Atoms, Neutron, Proton, Electron, Number of Atom etc. are common to both Physics and Chemistry.

In Physics we study the entire universe. It includes the study of planets, sun, moon and the study of stars, its distance from Earth, Mass, Rotations, Cosmic rays etc. We learn in Physics the household articles, its structure, sound instruments, light instruments. Machines in industries, medical equipments, electricity and its different production methods etc.

In chemistry we learn, molecules, metals, non-metals, chemical reactions, formula compounds of organic and inorganic, earth science etc. Methods of preparing medicines for our daily usage, fertilizers, dying materials, cement, sugar etc. are known from Chemistry. Processes to purify the Water and air is available in Chemistry.

The word science is derived from the Latin Word 'Scientia' which means 'knowledge'. Various scientists gave various explanations to science.

Science is the intellectual and practical activity encompassing the systematic study of the structure and behavior of the Physical and natural world through observation and experiment.

Science involves more than the gaining of knowledge. It is the systematic and organized inquiry into the natural world and its Phenomena.

Science is about gaining a deeper and often useful understanding of the world.

Science is Systematic knowledge of the Physical or material world gained through observation and experimentation.

1.2 Nature of Physical Science

1. The World is understandable

Science presumes that the things and events in the universe occur in consistent patterns that are comprehensible through careful, systematic study. Scientists believe that through the use of the intellect, and with the aids of instruments that extend the senses, people can discover patterns in all of nature.

2. Scientific ideas are subject to change

Science is a process for producing knowledge. The process depends both on making careful observations of Phenomena and on inventing theories out of those observations. Change in knowledge is inevitable because new observations may challenge prevailing theories.

3. Scientific knowledge is durable

Although scientists reject the notion of attaining absolute truth and accept some uncertainty as part of nature, most scientific knowledge is durable. The modification of ideas rather than their outright rejection, is the norm in science; powerful construct tend to survive and grow more precise and to become widely accepted.

4. Science cannot provide complete answers to all questions

There are many matters that cannot usually be examined in a scientific way. There are for instance, beliefs that the existence of super natural powers. In other cases, a scientific approach that may be valid is likely to be rejected as irrelevant by people who hold to certain beliefs (such as miracles, fortune-telling, astrology and superstition).

5. Scientific Inquiry

Fundamentally, the various scientific disciplines are alike in their reliance on evidence, the use of hypothesis and theories, the kinds of logic use and much more. Scientists differ one another in what phenomena they investigate and how in they go about their work. Still the exchange of techniques, information and concepts goes on all the time among scientists and there are common understandings among them about what constitutes an investigation that is scientifically valid.

Scientific inquiry is not easily described. There is no fixed steps that scientists always follow. No one path that leads them unerringly to scientific knowledge. There are certain features of science that leads to mode of inquiry. Those features are especially the work of processional

scientists. Everyone can think scientifically about many matters of interest in every day life.

6. Science Demands Evidence

Scientific claims is settled by referring to observations of phenomena. Hence scientists concentrate on getting accurate data. Such evidence is obtained in Natural or in laboratory. To make their observations scientists use their own senses and instruments. To reliance on evidence, great value is placed on the development of better instruments and techniques of observations and the finding of anyone investigator or groups are usually checked by others.

7. Science is a Blend of logic and imagination

Scientific findings must confirm to the principle of logical reasoning that is testing the validity of findings by applying certain criteria of inference, demonstration and common sense.

Scientists do not work only with data and well developed theories. First they form tentative hypothesis about the way things may be. Such hypothesis are widely used in science for choosing what data to pay attention and what additional data to seek.

Scientific concepts do not emerge automatically from the data or from analysis alone. Inventing hypothesis or theories to image how the world works and then post to the test of reality is as creative as writing poetry, composing music etc. Sometimes discoveries in science are made unexpectedly. But knowledge and creative insight are usually required to recognize the meaning of the unexpected. Aspects of data that have been ignored by one scientist may lead to new discoveries by another.

1.3 Scope of Physical Science

The scope of physical science is elaborate. Physics is the study of matter and energy at all scales where as chemistry is the study of matter and energy at the scale of atoms and molecules. We see in all our walks of life physical science influences. We depend on physical science for just about everything that make modern life possible. You could not turn on a light, make a phone call or use a computer without the discoveries in Physics and Chemistry. People depend on physical science to lead a comfortable and happy life persons having good physical science knowledge having lot of employment opportunities in the fields of Education, Production of household articles, Department of transport, Aeroplanes, Communication, Medicine, Physical Science laboratories, Research institutions and much more.

Physical science candidates can make is career as an Assistant Physical Scientist, Astronaut, Chemist, Atmospheric Scientist, materials scientist, Hydrologist, Quality manager, Geo Scientist, Technician, Physics teacher and Physical Science teacher.

Number of vacancies are available for new talented persons in the following companies :

- Tata
- UTC Aerospace
- Hindustan Lever
- Microsoft
- SanDisk
- Motorola
- Godrej Aerospace

1.4 Need and significance of teaching Physical Science

To teach any subject, it is the primary fact to know the teachers and students. What is the purpose of teaching and what are the need and uses of teaching.

Physical Science is a very important subject. The aims and objectives of Physical Science teaching have undergone an immense change in the computer era. In the growth and development of an individual and to apply knowledge and skills in the working place, we need physical Science. Any how the following points will tell us the need and importance of Physical science.

1. To know the world around us.
2. To enable the individual to meet the demands of daily life.
3. To understand the natural phenomena clearly.
4. To develop self-confidence, self-reliance and patience.
5. To apply physical concepts and principles in a new situations.
6. To apply physical science knowledge and skills in the working place.
7. To apply physical science knowledge in the other school subjects.
8. To find the alternate methods while solving the problem.
9. To develop the skill of handling laboratory instruments, observations and verifying the conclusions.

10. Providing useful work for leisure
11. To inspire and appreciate the scientists and their discoveries
12. To involve students in stimulating scientific thoughts

Significance of teaching Physical science

1. To understand the nature of Physical Science.

- To understand the scientific principles, laws and logic which is used in present or in future.
- To understand science in healthy and truth.
- To understand the world around us and the natural laws
- To understand the various parts of science and their relationship

2. To obtain thinking skills

When student sit in the class rooms or in laboratories to get Physical education knowledge, they obtain several skills.

- Obtain the problem solving skill
- Obtain drawing skills to be used in experiment
- Laboratory knowledge need in future

3. To obtain the observation skill

Only in Science subject five senses are used to find the truth. Concepts and principles are observed properly and find the truth with the help of experiments

4. To develop the scientific attitudes

Students get the following scientific attitudes while learning Physical Science

- Getting broad scientific attitudes
- To get the interest in Physical Science
- To obtain knowledge to solve the science problems
- Not to be believer of superstitious.
- To respect the views of others
- To be open minded

5. To get training in scientific methods

Students solve their day to day problems using the knowledge obtained in Scientific methods

Steps in Problem solving using scientific method

- Identify the problem
- Define the problem
- Collect required data to solve the problem
- Forming hypothesis
- Testing hypothesis
- Apply the results in various other situations

6. To create interest in Science

Teaching Science help students to participate in scientific activities with interest. Students take interest in participating scientific activities namely Science Club, Science Excursion, Field Trip and Science exhibition etc.

7. To know how physical science is used for technological development

We cannot attain technological development without the help of Physical science. The things which we are using to live comfortable are made with the help of Physical science. The basis for industrial development is Electricity. Physical science gives the various methods for preparing electricity. It gives various methods to keep environment without pollution. New technologies are introduced with the help of Physical Science in the fields of Computer Science, Communication and industrial development.

8. Training for better living

A science student should know the ways and means of prevention and eradication of diseases and to maintain good health. He should know how to adjust himself with his own domestic social environment and economic and cultural conditions.

9. To make our leisure time usefully

Physical science make our leisure time usually by repairing household articles, photography, two wheeler repairing etc. From these activities we understood the scientific concepts clearly and it also gives some remuneration. The aims and objectives of education should also give how to make our leisure of time usefully.

10. To prepare students for the future

Science helps to prepare students for their future. In Higher secondary stage, students should select the subjects according to their choice and keeping in mind their future. School should encourage students to select subject to study engineering, technical or communication branches for their

future. Apart from physics and chemistry school should encourage students to take new subjects like Bio Physics, Astro Physics, Computer Science, Bio Chemistry etc. Students who are studying physical science in +2 stage will study these courses and make their future bright.

1.5 Values of teaching Physical Science

Importance and functions of values

Values are general principles to regulate our day to day behaviour. They are not only gives direction to our behaviour but also ideals and objectives in themselves. Values deal what ought to be. Our values are the basis of our judgements about what is desirable, proper, correct, important, worthwhile and good. It controls our disruptive behaviours; values enable individuals to feel that they are above us. Live with the following objective.

"Let Noble thoughts come to me from all directions"

Values of teaching Physical science

The real importance of the subject in our modern life can be quite obvious from its chief values which are described below.

1. Practical Value

Practical value deals the relationship between Physical science and day to day activities of our life. Utilization of the various facts drawn from the study of Physical science in modern life has revolutionized our life. Today we cannot find even a single thing which is left untouched by the hands of physical science.

There is no life activities without physical Science. To lead a healthy and happy life, the new things discovered

Aims and objectives of teaching Physical Science

namely various kinds of Fans, Air coolers, Air conditioners, Vehicles instruments used in Kitchen. Office usage things like computer and other articles, industrial equipments are all made with the help of Physical Science.

2. Social Value

Social values deals with how physical science is used in maintaining and administering general social and economic life of his country. To administer social industries like Central and State Government administration, cooperative societies, Banks, Insurance companies, industries etc. physical science knowledge is needed to make decisions in human affairs and official affairs. Physical Science methods such as scientific method, laboratory method, logical method, deductive method etc. are used to analyse, interpret social values. Today the status of a hospital is depend on the latest physical science equipments available there. To identify the physical science equipments available there. To identify the disease and to give treatment Physical science equipments are largely used in hospitals. From the blood test to identify and cure cancer hospital depend on physical science equipments. Today we make electricity from the municipal waste. The contribution of physical science is great in the modern society. Through the study of Physical science we get self confidence, patients, strong determination systematic work, helping others etc. and all these qualities leads to live a satisfactory life.

3. Disciplinary Value

Disciplinary values deals with how physical science is used to do any work with self confidence and discipline. The learning of Physical science involves some scientific disciplines and scientific attitudes which are transferable to our later life also. It involves self-expression, creativeness

open mindedness, critical thinking and observation suspended judgment.

4. Cultural Value

Cultural values deals with how physical science is used in the development of cultural and civilizational values of a nation. The role of Physical science in the development of modern civilization can be quite obvious just by our comparison with our ancestors. Our present culture and advancement in our standard of living gives a clear cut picture of our cultural development and role of physical science in this field for removing old traditional beliefs and superstitions.

5. Moral Values

Moral values are the standards of good and evil which governs an individuals behaviour. Science has more moral values. Science is the search for truth in a faithful manner. Promotion of science along with the growth of moral values in necessary for human development. Scientists strongly abhor fraud, error and pseudoscience while they value reliability, testability, accuracy, precision and simplicity. These values are generally derived from our experience in research. Good views of others are accepted by scientists. For the welfare of mankind scientists conduct research and invent new things. It teaches the pupil to be intellectually honest and truthful in our daily activities.

1.6 Aims and objectives of teaching Physical Science in schools

Aims:

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the uses of teaching. We use the term 'aims' to mean the broad goals which our educational system (the

whole of the Indian society involved) embraces and which we are expected to attain. The aims of education are based on philosophical and socio-psychological aspects of society and culture. Aims are long term goals.

Objectives:

Objectives are the milestones to reach the destination i.e. to attain the aim or ultimate goal of education. Objectives are specific, direct and practical in nature; objectives are related to the learning outcomes or change in the behaviour of the students. The objectives are meant for both the teacher and the students. Even at the planning stage i.e. before entering the classroom the teacher asks himself, "what changes in behaviour of the students can I bring about through this lesson?" such changes constitute his (teacher's) educational objectives. Objectives are short term goals.

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the need and uses of teaching. Physical Science is the queen of all sciences. The aims and objectives of Physical Science teaching has undergone an immense change in the computer era. In the growth and development of an individual and to apply Physical Science knowledge and skills in the working place, we need Physical Science. We can not give the importance of Physical Science in a few lines. Any how the following points will tell us the need and importance of Physical Science and the aims and objectives of Physical Science.

i. Knowledge objectives

1. To know the world around us
2. To understand the natural phenomena clearly

3. To enable the individual to meet the demands of daily life.
4. To understand the culture, social development and civilization.
5. Perform Physical Science operations and manipulations with confidence speed and accuracy.
6. To collect, organize, analyze and interpret data and make conclusions and predictions from its results.
7. To prepare students to college level

ii. Skill Objectives

1. To develop skill in the use of Physical Science tables, statistical tables and computers.
2. To apply Physical Science concepts and principle in a new situations.
3. To apply Physical Science knowledge and skills in the working place.
4. To apply Physical Science knowledge in the other school subjects.
5. To develop the skill to express Physical Science thoughts clearly and accurately.
6. To Develop Physical Science curiosity and use inductive and deductive reasoning when solving problems.
7. To recognize patterns
8. The real aim of teaching Physical Science is to develop power, not knowledge. The knowledge acquired through reasoning becomes power, which helps in discovering new facts.
9. To develop logical and critical thinking

10. To develop various skills like accuracy, neatness in drawing geometrical figures.

iii. Appreciation Objectives

1. To recognize that Physical Science is a powerful and fundamental tool of science with limitless applications.
2. To enable the pupils to appreciate the subject of Physical Science in Engineering and technology.
3. To enjoy Physical Science and develop patience and persistence when solving problems.
4. To exhibit how Physical Science is related with other school subjects.
5. To appreciate the culture values of Physical Science

iv. Attitude Objectives

1. To develop self confidence, self reliance & patience
2. Develop a positive attitude towards learning Physical Science.
3. Think precisely, logically and critically in any situation.
4. To inculcate the habit of hardwork, patience, self confidence, regularity etc.
5. To develop moral values of the pupil

v. Social Objectives

1. All round development of the student for improvement of his personality
2. To develop willingness to work collaborately.
3. To develop the leadership qualities.
4. To develop the habit of helping others.
5. To develop self-confidence, self-reliance & patience.

1.6.1. Aims and objectives of teaching Physical science in Schools.

It can be divided into three stages

1. Aims and objectives of teaching Physical science at Primary stage.
2. Aims and objectives of teaching Physical science at High School stage.
3. Aims and objectives of teaching Physical science at Higher Secondary stage.

1. Aims and objectives of teaching Physical science at the Primary Stage

- To create interest in learning Physical Science.
- To relate Physical Science with day to day activities
- To increase the skill of observation
- To create good habits and to lead a healthy life with the help of environment.
- To stimulate the scientific attitude.
- To know the basic truth of the things he see in his environment.
- To make understand various principles of Physical Science with the help of assignments and projects.
- To develop good character and Patriotism

2. Aims and objectives of teaching Physical science at the high school level

i. Students acquire the knowledge of physical science

- To explain fundamentals Principles and concepts with the help of day to day activities.
- Facts for scientific study
- Inter dependence and relationship of different branches of science
- Natural Phenomena going on
- Knowledge of general rules of health

ii. Skills

Science students should acquire skills in experimentation, construction, observation and drawings.

Laboratory work in science infused with the spirit of inquiry, provides students with hands-on experience and develops a scientific attitude.

iii. Abilities

- Sense a problem
- Analysis the problem
- Finding solution to the problem
- Organise exhibitions, excursions and fairs
- Discuss and express scientific terminology.

iv. **Attitudes**

- Critical in observation and thought
- Open-minded
- Respecting others' views and discuss his problems with others
- Not a believer of superstitious
- In search of answers to 'what' or 'why' and 'How' of the things he observes.

v. **Interest**

- To create interest in reading scientific literature
- To participate with interest in science club, science exhibition, field trips etc.

3. **Aims and objectives of teaching Physical science at the Higher Secondary level**

- To familiarize the pupil with the world in which he lives and to make him understand the impact of science on society so as to enable him adjust himself to his environment.
- To develop the scientific attitude
 - Followers to cause and effect relationship
 - Adopts planned procedure in solving a problem
 - Truthful in his experimentation and conclusion
- To understand the scientific development
- To acquire the knowledge of scientific methods,

1.6.2 **Difference between Aims and Objectives**

S. No.	Aims	Objectives
1	Aims are long term goals	Objectives are short term goals
2	Aims are broad and common	Objectives are narrow and specific
3	Philosophy, Sociology is main source of aim	Psychology is the main source of objectives
4	They are difficult to achieve	they can be achieved conveniently
5	These can't be evaluated	They can be evaluated
6	School, Society and Nation is responsible to achieve them	Generally teacher is only responsible
7	Aims are non-observable	Objectives are observable behaviours
8	To achieve aims we have to go beyond curriculum	Objectives are achievable within the curriculum

1.7 **Instructional Objectives and Behavioural Objectives****Instructional Objectives**

Instructional objectives are basically statement which clearly describe an anticipated learning outcome.

Instructional objectives are helpful to the teacher as well as the learner throughout the learning process and are invaluable in the evaluation process. An instructional objective is the focal point of a lesson plan. Before start teaching teacher must be guided by instructional objectives. Instructional objectives can be classified into three domains of learning i.e. cognitive, affective and psychomotor. Objectives should state what the learners should be able to do after the instruction is complete. The process of how the instruction happens is not considered as an objective.

Purpose of Writing Instructional Objectives

- Define the desired outcome of instruction in terms of tasks that student will be able to perform.
- Establish expectation for the student.
- Focus course development and teaching activities on relevant information and skills.
- Provide a clear reason for teaching
- Provide criteria for eliminating unnecessary information and activities
- Indicate how success can be measured

Characteristics of Instructional Objectives

Instructional objectives are specific, measurable, short term, observable student behavior. They indicate the desirable knowledge, skills of attitudes to be gained at the end of teaching. It describe the performance of a learner should be able to demonstrate after instruction.

- A well-written objective should describe a learning outcome. It should not describe a learning activity
- An objective focuses on the learner, not on the teacher. It describes what the learner will be expected

to be able to do. It should not describe a teacher activity.

- If an instructional objective is not observable, it leads to unclear expectations and it will be difficult to determine. The key to writing observable objectives is to use verbs that are observable and lead to a well defined product. Verbs such as "to know" to understand, "to enjoy" to appreciate etc. are vague and not observable. Verbs such as "to identify", "to list", "to compute" are explicit and describe observable actions or actions that lead to observable products.

- There are many skills that cannot be directly observed. The thinking process of a student to solve a mathematics problem cannot be easily observed. However one can look at the answers or the steps to arrive at an answer.
- Objective to be sequentially appropriate
- Objective to be attainable within reasonable amount of time.
- Objective to be developmentally appropriate (within students level of intellectual)

Importance of Instructional Objectives

Instructional objectives are important to both learners and instructors. They help learners plan their study and prepare for examinations. They guide the instructors in planning instruction and devising tests. Instructional objectives play an important role in the teaching and learning process. They are the heart of teaching.

1. To provide direction for the instructional process by clearly stating the intended learning outcomes.

2. To convey instructional intent to students, parents and educational authorities.
3. To provide a basis for evaluating student's learning by describing the performance to be measured.
4. They state in clear terms behavioural changes expected from a learner after with a learning activity.
5. Without instructional objectives, it will be very difficult for a teacher to access the behaviours of the students in the three domains of education. (Cognitive, affective and Psychomotor Domains)
6. Instructional objectives encourage reflection and good course design and development.
7. Instructional objectives help to define reinforcement situations of the learners.
8. They enable the teacher to modify methods to suit the process of eliciting those behaviours that the objectives specify.
9. They help the teacher to know how well the students have covered the content.
10. They give direction to both teacher and students in the selection and use of materials, methods and activities for learning.

Behavioral objectives

A behavioral objective is a learning outcome stated in measurable terms, which gives direction to the learner's experience and becomes the basis for student evaluation.

Points in writing behavioral objectives

1. Begin each behavioural objectives with a verb. The critical aspect of any behavior is the verb selected from learning activities.

2. State each objective in terms of learner performance. A behavioral objective is one that is considered to be observable and measurable. Behaviour is generally constructed to be an action of an individual that can be seen, felt or heard by another person.
3. State each objective so that it includes only one general learning outcome.

Examples of Behavioral Objectives

Knowledge	:	Remembering previously learned facts
Comprehension	:	Ability to understand or grasp the meaning of material.
Application	:	Ability to use previously learned material in new and concrete situations.

Motivation and Behavioural Learning

Motivation plays an important role in behavioural learning. Positive reinforcement can be motivators for students. For example, a student who receive praise for a good test score is much more likely to learn the answers effectively than a student who receives no praise for a good test score. The student who receives no praise is experiencing negative reinforcement - their brains tell them that though they got a good grade, it did not really matter, so the material of the test becomes unimportant to them. Conversely students who receive positive reinforcement see a direct correlation based on that response to a positive stimulus.

Purpose of Behavioural objectives

- Guide for the teacher to design of instruction.
- Guide for the teacher for evaluation

- Guide for the learner relative to learning focus
- Guide for the learner relative to self assessment
- Makes teaching more directed and organized
- Forces teacher to think carefully about what is important.
- Helps teacher to avoid unnecessary repetitions in teaching.
- Provides visibility and accountability of decisions made by teachers and learners
- Helps students make decisions regarding prioritizing
- Helps relationship between teacher and learner.

Examples of Behavioural objectives

General Objectives : The learner will be able to know the knowledge of measurements.

Specific Objectives : The learner will be able to use Vernier Caliper to find the length of an object.

1.7.1 Writing Instructional Objectives in terms of Behavioural Objectives

A behavioral objective is a way by describing the objectives of a content in terms of what the learner should be able to do at the end of the lesson. A behavioural objective must be stated clearly and precisely so that everyone who reads it will know exactly the desired outcome of the content.

In writing a behavioural objective, it is important to include the verb that depicts what the learner is expected to do, i.e. it should be measurable.

Examples

- 1) The students will be able to describe the process of transpiration in plants

Here the students are going to describe at the end of instruction. Here the emphasis is on learners and the learning outcome. Here learning is a change of behavior.

- 2) To identify proper fractions

Here the learner and his performance are given.

- 3) The students will understand the three laws of reflection

Here the verb understand does not indicate behavior that can be observed directly. It does not indicate any specific behavior. If the objective is written as.

The students will tell the three laws of reflection.

Then it is observable behavior. Hence the verb 'understand' can further be explained and classified by listing under that some specific behavior like tell, identify, describe, give, differentiate etc. In other words the verb understand is more general and the verb tell is more specific as compared to other.

Let us discuss the two objective in details

- i. The pupil will know the working of a steam engine.
- ii. The pupil will be able to describe in about ten to fifteen sentences the working of a steam engine.

In the above example i) one cannot see other's mind to determine what he knows. We are not able to observe some aspect of this behavior that corresponds to his knowledge. Hence an objective should indicate the kind of performance which will be accepted as evidence.

Thus in the second example, well-stated instructional objectives are given in observable behavioral terms.

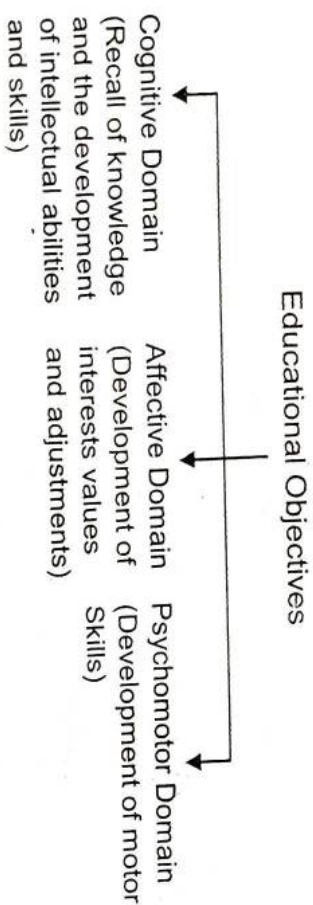
Few examples of well-stated Instructional Objectives

1. The students will be able to describe the three causes of World War II as discussed in the class.
2. The class will be able to identify correctly, all the parts of the hibiscus flower.
3. Students will tell the name of any four corporations in Tamil Nadu.
4. Students will be able to identify the part of a simple sentence.
5. Students draw a right angled triangle with given two sides.

1.8 Blooms Taxonomy of Educational Objectives

To define understand and classify the educational objectives American educationalist Benjamin S. Bloom and his associates at the University of Chicago divide it into three domains i.e. cognitive, affective and psycho-motor and explains clearly to the teachers.

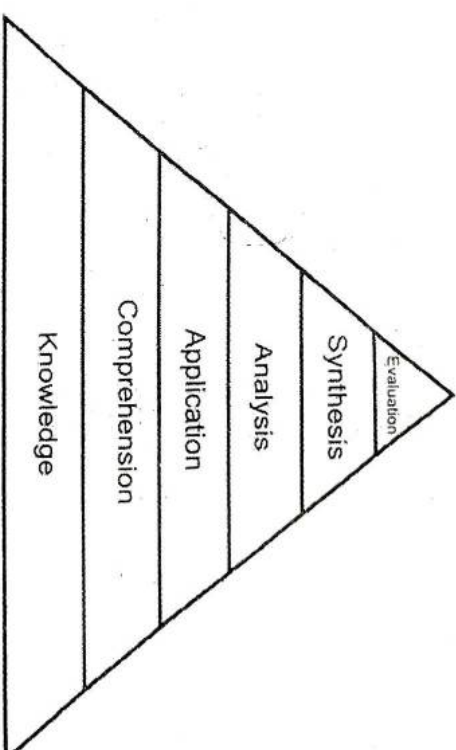
A brief account about these is given below.



In each domain, objectives are hierarchical order starting from easy to difficult. Hence this arrangement is called "Taxonomy of Educational objectives.

1.8.1 Cognitive Domain

Cognitive domain includes those objectives which deal the recall or recognition of knowledge and the development of intellectual abilities and skills. This domain consist of six parts and arranged in a hierarchical order, in increasing levels of difficulty (from lower to upper)



Knowledge forms the basis for all cognitive objectives and all other objectives are not possible without knowledge. Similarly if one is able to apply knowledge in a new situation, it can be inferred that he has to achieve lower levels of objectives namely knowledge and comprehension clearly. Thus a higher level objectives subsumes to lower ones.

While teaching mathematics for Secondary and Higher Secondary students give priority to the first three objectives of cognitive domain namely knowledge, comprehension and application. The remaining three objectives Analysis, Synthesis and Evaluation may be trained to them according to their maturity of mind.

Knowledge :

This objective is to remember and recall or recognize the already learnt information and acquire more and more knowledge.

Comprehension :

Objective at this level requires learners to understand the knowledge without any doubt. He should have ability to arrange data in order.

Application :

This objectives requires students to apply already learnt knowledge to a new and unfamiliar situations. Think critically in depth.

Analysis :

Objective at this level requires the learners, to attain the ability to break up a given communication into parts to understand it better.

Synthesis :

Synthesis is just opposite to Analysis. The ability to draw upon elements from various sources and put them together to produce a new structure. This objective gives divergent thinking and creativity.

Evaluation :

This is a highest objective of cognitive domain. The ability to make judgements about something for given purpose. Judgement can be both quantitative or qualitative.

It helps teachers in the following ways:

- Helps to assess and judge the difficulty of assignments.
- It increase the accuracy of the lesson
- Simplifies action to ease personal learning
- Designed a combined assessment
- Helps for project based plans for learning
- Helps for group discussion

Instructional objects of Cognitive Domain

Parts of Cognitive Domain	Instructional Objectives
Knowledge Student knows the physical science symbols, concepts, definitions, formulas, statements etc. of physical science subject	Student 1. Recalls the physical science concepts and statements. 2. Recognises what he learned.
Comprehension Student understand the physical science symbols, concepts, definitions, formulas, statements	Student 1. Gives examples 2. Give reasons 3. Compares 4. Differentiates 5. classifies 6. relates 7. detect errors and rectify 8. using symbols 9. Explains 10. Evaluate the result
Application Using physical science knowledge and comprehension to solve the problem in a new situation.	Student 1. Examine what is given in the problem? What is need? 2. Examine the sufficiency of the data. 3. Select appropriate formula 4. Solve the problem in an appropriate method 5. Applying alternate method 6. Verifying the result.

Parts of Cognitive Domain	Instructional Objectives
Analysis Break up a given physical science statement into small meaningful parts	Student 1. Understand the meaning of the hypothesis 2. From the given data draw conclusion 3. Identify motives / causes for the physical science concepts. 4. Discriminates, selects, illustrate physical science concepts.
Synthesis Putting elements from various sources and put them together to produce a new structure	Student 1. Design a method to perform a new task. 2. Revises the process to improve the outcome. 3. Compile, generate, create Science concepts. 4. Solves the problem 5. Develop the idea in advance.
Evaluation Evaluating Qualitative or quantitative judgement for physical science elements or methods.	Student 1. Evaluate the solution of a given problem 2. Select the most effective solution. 3. Explain and justify a new proposal 4. Assess / defend physical science concepts. 5. Review / Criticise physical science statements.

1.8.2 Affective Domain

The affective domain refers to emotional and attitudinal engagement with the subject matter while the cognitive domain refers to knowledge and intellectual skills related to the subject matter.

Affective domain includes the development of feelings of the educational and social attitudes and to increase the emotional objectives namely attitude, interest, values and appreciation. The different objectives under affective domain are Receiving, Responding, Valuing, organisation and characterisation.

1. Receiving:

This is the first step for the learner to learn what the teacher teaches. Listen attentively to a particular phenomena or stimuli.

2. Responding:

This is also called interest objective. Pupil is sufficiently motivated to attend actively. Interest in Physical Science lends the student to select a particular vocation throughout his life. Interest can be developed in solving puzzles, reading mathematical books and journals etc.

3. Appreciation :

Student appreciates Physical Science for its use in daily life and how it helps to other subjects.

4. Organisation :

This objective brings together different values, resolving conflicts among them, starting to build an internally consistent value system.

5. Valuing:

It means individual's own valuing or assessment of thing, phenomena or behaviour. It is also called attitude.

Development of proper attitude in Physical Science is one of the major objectives of teaching Physical Science.

Instructional Objectives of Affective Domain

Parts of Affective Domain	Instructional Objectives
Receiving Students listens attentively what the teacher teach.	Student 1. Identifies 2. Asks 3. Understand 4. Points out errors boldly. 5. Listen to others attentively.
Responding Student develops interest in science. Student participate in class activities.	Student 1. Reads books of science 2. Solves science puzzles 3. Writes activities on science for school magazine. 4. Gives short cuts for solving problems. 5. Does problems apart from syllabus. 6. Gains science knowledge more than the syllabus. 7. Participates in science club activities. 8. Participate in class discussions.
Appreciation Student appreciates science for its use in daily life.	Student 1. Solving the problems of other subjects using science. 2. Appreciates how science is useful in his daily life.
Organisation Student organizes values into priorities by contrasting different	Student 1. Recognize the need for balance between freedom and responsible behavior

Parts of Affective Domain	Instructional Objectives
values, resolving conflicts between them.	2. Explain the role of systematic planning in solving the problems. 3. It becomes habit to think logically.
Valuing Student attaches to a particular object, phenomena or behaviour	Student 1. Shows self reliance when works independently 2. Co-operates in group activities. 3. Revises judgements and changes behavior in light of new evidence. 4. Values people for what they are not how they look. 5. Accepted statements when logically proved.
	Student 1. Identifies 2. Asks 3. Understand 4. Points out errors boldly. 5. Listen to others attentively.
	Student 1. Reads books of Science 2. Solves Science puzzles 3. Writes activities on Science for school magazine. 4. Gives short cuts for solving problems. 5. Does problems apart from syllabus. 6. Gains mathematics knowledge more than the syllabus.

Parts of Affective Domain	Instructional Objectives
	- Participates in Science club activities. - Participate in class discussions.
	Student - Solving the problems of other subjects using mathematics. - Appreciates the symmetrical pattern of figures seen in his daily life. - Appreciates how Science is useful in his daily life.
	Student - Recognize the need for balance between freedom and responsible behavior - Explain the role of systematic planning in solving the problems. - It becomes habit to think logically.
	Student - Shows self reliance when works independently - Co-operates in group activities. - Revises judgements and changes behavior in light of new evidence. - Values people for what they are not how they look. - Accepted statements when logically proved.

1.8.3 Psychomotor Domain

Elizebath Simpson built her taxonomy on the work of Bloom and others. Simpson's Psychomotor domain is comprised of utilizing motor skills and coordinating them. Simpson's taxonomy has a focus toward the progression of mastery of a skill from observation to invention.

The psychomotor domain includes Physical movement, coordination and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, procedures or techniques in execution. The seven major categories are perception, set, Guided Responses mechanism, complex over to Response, Adaptation and Origination.

The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1. Perception

It means observing the act of the teacher.

2. Set

Readiness to act. It includes mental, physical and emotional sets

3. Guided Response

Following directions of the teacher and differentiating among various movements. This is the early stage in learning complex skill.

4. Mechanism

This is the intermediate step in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.

5. Complex Overt Response

This is the highest level of proficiency and it becomes natural and automatic. Proficiency is indicated by a quick, accurate and highly coordinated performance, requiring a minimum energy.

6. Adaptation

Skills are well developed and the individual can modify movement patterns to fit special requirements

7. Origination

Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.

Instructional Objectives of Psychomotor Domain

Parts of Psychomotor Domain	Instructional Objectives
Perception The ability to use sensory cues to guide motor activity	Student 1. Detects non verbal communication cues 2. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball. 3. Observing the act of the teacher.
Set Readiness to act. It includes mental, physical and emotional sets.	Student 1. Repeat the act of the teacher. 2. Acts upon a sequence of steps to solve a problem. 3. Shows desire to learn a new process.

Parts of Psychomotor Domain	Instructional Objectives
Guided Response Following directions of the teacher and differentiating among various movements. This is the early stage in learning complex skill.	Student 1. follows 2. react 3. response 4. copies 5. performs
Mechanism This is the intermediate step in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.	Student 1. Drives a cycle 2. assembles 3. dismantles 4. measures 5. mixes
Complex Overt Response This is the highest level of proficiency and it becomes natural and automatic. Proficiency is indicated by a quick, accurate and highly coordinated performance, requiring a minimum energy.	Student 1. Operates a Computer quickly and accurately 2. assembles 3. builds 4. measures 5. organizes
Adaptation Skills are well developed and the individual can modify movement patterns to fit special requirements.	Student 1. Responds effectively to unexpected situations. 2. Modifies method to meet the needs. 3. Perform a task with a new method.
Origination Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.	Student 1. Constructs a new theory. 2. designs 3. creates 4. builds 5. composes

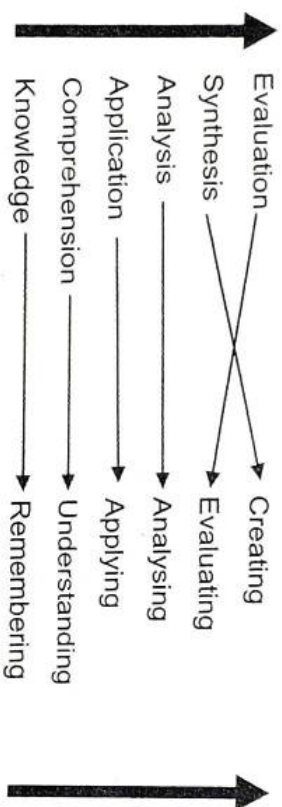
The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1.9 Revised Bloom's Taxonomy

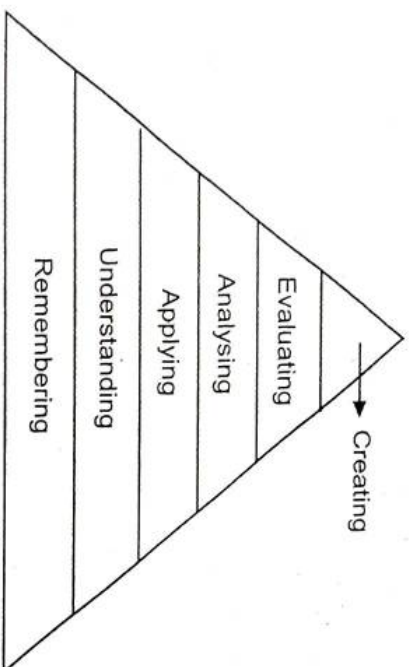
In 1990 Loris Anderson former student of Bloom revised the taxonomy and made a number of changes.

Bloom's Taxonomy

Revised Bloom's Taxonomy



Revised Bloom's Taxonomy



Remembering:

The knowledge category was renamed as remembering. Knowledge is an outcome or product of thinking and not a

form of thinking. Hence the word knowledge was inappropriate to describe a category of thinking and was replaced with the word remembering.

Remembering is recalling information. This objective is to recognizing, listing, describing, naming, finding already learnt information.

Understanding:

Objective at this level requires learners to understand the knowledge without any doubt. He should have knowledge to interpret, explain, classify and summarize.

Applying:

Objective at this level requires the learners to apply information in another familiar situation and executing in a best way.

Analysing:

Objective at this level requires the learners to attain the ability to break up a given information into parts to explore understanding and relationships between them.

Evaluating:

The ability to make judgements about a decision or course of action. Judgement can be both quantitative or qualitative.

Creating:

By putting elements together from various sources to form the ability to generate new ideas, products or ways of viewing thing in a different way. Designing, planning, producing, inventing new things and objects is the objective at this level.

Potential activities of cognitive domain

Remembering:

1. Make a list of the main events of today.
2. Write Boyle's Law.
3. Objective questions are used to measure the learner's skill.

Understanding:

1. Retell the story in your own words.
2. Prepare a flow chart to illustrate the sequence of events.
3. Comparing the concepts

Applying:

1. Construct a model to demonstrate how it works.
2. Make a clay model to explain a science concept.
3. Solving or explaining the problem

Analyzing:

1. Design a questionnaire to gather information.
2. Construct a graph to illustrate given information.
3. Here thinking skill is developed

Evaluating:

1. Conduct a debate about an issue of
2. Prepare a case to present your view about.....
3. Using different ideas, student understand the concepts

Creating:

1. Invent a machine to do a specific task.
2. Design a magazine cover for sports day function.
3. Developing skill to create a new hypothesis.

1.10 Writing instructional objectives for different Physical Science Topics

To write objectives for a particular physical science topic, begin with an action verb while writing objectives. Do not use verbs such as learn, understand, Discuss, explain etc, as these do not specify exactly how students will demonstrate their knowledge. Use specific action verbs that will demonstrate their exactly what the student will be doing during the lesson.

Remember that your objectives must be observable. Can you see a student determine? Can you see a student understand? Do not use unobservable behaviors as objectives.

Physics

Std : 9

Topic : Boyle's law

Instructional Objectives of Teaching

1. Knows the Boyle's law
2. Understand the Boyle's law
3. Using knowledge and understanding solves – problem in a new situation.
4. Solves the problem with interest
5. Writes oral information into symbols
6. Verifies Boyle's law in the laboratory

Chemistry

Std : 9

Topic : Common properties of Covalent Compounds

Instructional Objectives of Teaching

1. Knows the Common properties of Covalent compounds
2. Understand the common properties of covalent compounds.
3. Using knowledge and understanding verifies the common properties of covalent compounds using simple experiments.
4. Doing the experiment with interest
5. Understand the difference between ionic compounds and covalent compounds.

Std: 6 Topic: Measurements - Physical Quantities: Length

Learning Objectives

Students will be able to

- Knows what measurement is
- Understand fundamental units and derived units
- Learn the units for various physical quantities
- Identify zero error and parallel error.
- Apply the concept of measurement in day-to-day activities
- Evaluate the value of physical quantities.